CBSE Practice Worksheet

Class 6 | Science | Difficulty: Hard | Set C

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. What is the magnetic quantum number for 3d orbital?
   (A) -2 to +2
   (B) -3 to +3
   (C) -1 to +1
   (D) 0 only
2. What is the wavelength of Lyman series limit?
   (A) 91.2 nm
   (B) 121.6 nm
   (C) 656.3 nm
   (D) 364.6 nm
3. What is the escape velocity from Earth's surface?
   (A) 8 km/s
   (B) 11.2 km/s
   (C) 15 km/s
   (D) 25 km/s
4. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
   (A) 25%
   (B) 33%
   (C) 50%
   (D) 75%
5. What is the oxidation state of Mn in $KMnO_4$?
   (A) +2
   (B) +4
   (C) +6
   (D) +7
6. What is the coordination number in FCC structure?
   (A) 6
   (B) 8
   (C) 12
   (D) 14
7. Which type of mutation involves change in single nucleotide?
   (A) Deletion
   (B) Insertion
   (C) Point mutation
   (D) Frameshift
8. Which principle explains the working of a hydraulic lift?
   (A) Archimedes
   (B) Pascal
   (C) Bernoulli

(D) Newton

9. What is the wavelength of de Broglie wave for an electron?
   (A) $\lambda = h/mv$
   (B) $\lambda = mv/h$
   (C) $\lambda = hv/m$
   (D) $\lambda = m/hv$

10. Which enzyme is responsible for DNA replication?
    (A) DNA ligase
    (B) DNA polymerase
    (C) Helicase
    (D) Primase

Section B: Fill in the Blanks (1 mark each)

11. In SN2 reactions, the rate depends on both ___ and nucleophile.
12. The enzyme that converts prothrombin to thrombin is ___.
13. The Krebs cycle occurs in the ___ of mitochondria.
14. The band gap in semiconductors is typically ___ eV.
15. The process of splitting water during photosynthesis is called ___.
16. Raoult's law applies to ___ solutions.
17. The uncertainty principle was given by ___.
18. For a reversible process, entropy change of the universe is ___.

Section C: Short Answer Questions (2 marks each)

19. Describe the light-dependent and light-independent reactions of photosynthesis.

20. Derive the lens formula and explain its applications.

21. Explain Le Chatelier's principle with examples.

22. Explain the structure and working of a nuclear reactor.

23. Describe the mechanism of nerve impulse transmission.

24. Explain hybridization with examples of sp, sp^2, and sp^3.

25. Describe electromagnetic induction and derive Faraday's law.

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